Wage Determination and Inequality using supervised and unsupervised learning methods

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1 Introduction

Income inequality is an important topic in economics and public policy. In many countries, the gap between low-income and high-income individuals has increased over time. Previous research shows that income differences are strongly related to education, working hours, and access to capital income. These factors play a key role in explaining why some individuals earn much more than others (Ganaie et al., 2018).

In recent years, large datasets have become more available. This has led to a growing use of statistical learning and machine learning methods in income analysis. Supervised learning methods are often used to classify individuals into income groups, for example earning more or less than $50,000 per year. Many studies show that logistic regression is a strong and reliable method for income classification. It offers good prediction results and is easy to interpret compared to more complex models (Zaid and Rajendran, 2023; Yamnampet, 2017).

Unsupervised learning methods are also useful in income studies. Methods such as k-means clustering help identify groups of individuals with similar socio-economic characteristics. These methods do not use income during the grouping process. Instead, they help explore hidden patterns and differences

in the data. Clustering is mainly used for exploration, not for prediction (Warchalska-Troll, 2022).

This study applies both supervised and unsupervised learning methods to the Adult Income dataset from the U.S. Census. The goal is not to find causal effects, but to analyze income inequality using prediction and exploration. Supervised learning is used to predict whether an individual earns more than $50,000 per year. Unsupervised learning is used to identify socio-economic profiles based on education, working hours, and capital income.

Overall, this project shows how different statistical learning methods can be used together to study income inequality. The focus is on clear interpretation, model comparison, and careful discussion of results. The study provides a simple and transparent example of applied data analysis in economics and social sciences.

2 Data Description

2.1 Data Source

The analysis in this project relies on the Adult Income dataset, which is derived from the 1994 U.S. Census Bureau database and distributed via the UCI Machine Learning Repository. This dataset was originally prepared by Becker and Kohavi for research in income classification and has since become a benchmark dataset in applied statistics and machine learning.

Each observation corresponds to an individual adult living in the United States. The dataset provides a broad set of socio-economic characteristics, covering demographic attributes, educational attainment, marital status, occupational categories, labor supply indicators such as weekly working hours, as well as information on capital gains and losses and country of origin. These variables allow for a detailed examination of heterogeneity in income outcomes across individuals.

In its raw form, the dataset contains 48,842 observations, split into training and test samples. As is common practice with this dataset, records containing ambiguous or missing information were excluded during preprocessing. The final dataset used for the empirical analysis therefore consists of 30,162 individuals and 15 variables. The data are purely cross-sectional and represent a snapshot of income and individual characteristics at a given point in time.

2.2 Outcome Variables

The primary outcome variable considered in this analysis is individual income, captured by a binary indicator denoted as *income*. This variable classifies individuals into two categories depending on whether their annual income exceeds a threshold of $50,000. Specifically, the outcome takes the value *">50K"* for individuals earning more than $50,000 per year and *"≤50K"* otherwise.

This binary income measure serves as a practical proxy for distinguishing between lower- and higher-income individuals and is commonly used in empirical studies of wage determination and income inequality. Rather than modeling income as a continuous variable, the dichotomous outcome allows the analysis to focus on relative income positions and facilitates the application of classification-based statistical learning methods.

In the cleaned dataset used for analysis, approximately 25 percent of individuals fall into the higher-income category, while the remaining 75 percent earn $50,000 or less per year. This class imbalance is a relevant feature of the data and is taken into account in both descriptive analysis and model evaluation. The income indicator is treated as the central variable of interest throughout the study and provides the basis for exploratory comparisons, hypothesis testing, and supervised learning exercises.

2.3 Explanatory Variables

The dataset contains a set of explanatory variables describing individual socio-economic characteristics. These variables are used to analyze differences in income outcomes and are grouped below for clarity.

Demographic variables include age, sex, race, marital_status, and native_country. These variables describe basic individual characteristics and allow for comparisons of income across population groups.

Education-related variables include education and education_num, which capture the individual's highest level of educational attainment in categorical and numerical form.

Labor market characteristics are described by workclass, occupation, relationship, and hours_per_week. These variables provide information on employment type, occupational category, household role, and labor supply.

Income-related variables include capital_gain and capital_loss, which capture non-wage income components that may contribute to overall income differences.

Together, these explanatory variables provide a detailed description of individual characteristics that are relevant for studying income inequality and wage determination.

2.4 Data Preprocessing

All preprocessing steps were performed in R in a fully reproducible way, without manually modifying the raw data file. First, the dataset was imported and the official variable names were assigned (*age, workclass, fnlwgt, education, education_num, marital_status, occupation, relationship, race, sex, capital_gain, capital_loss, hours_per_week, native_country, income*).
Second, the categorical (text) variables were cleaned to ensure consistent formatting. Leading and trailing spaces were removed from character values, and ambiguous entries coded as `"?"` were treated as missing values and converted to `NA`.
Third, missing values were summarized across variables. Since missingness occurred in *workclass*, *occupation*, and *native_country*, a complete-case dataset was created by removing observations with at least one missing value. This cleaned dataset contains 30,162 observations and 15 variables and is used for the main empirical analysis.
Finally, categorical variables (*workclass, education, marital_status, occupation, relationship, race, sex, native_country,* and *income*) were converted to factors to ensure that they are correctly treated as categorical variables in tables, visualizations, and statistical models.

2.5 Descriptive Statistics

Table 1 reports descriptive statistics for selected numerical variables from the cleaned dataset used in the empirical analysis. The sample includes 30,162 observations after removing records with missing values in *workclass*, *occupation*, and *native_country*.
The statistics show substantial heterogeneity across individuals. In particular, hours_per_week varies widely, while capital_gain and capital_loss exhibit strong dispersion, with most observations taking the value zero but a small number reaching very large values. These descriptive patterns motivate the subsequent exploratory analysis and modeling exercises.

Table 1:Descriptive statistics for selected Variables

Variable	Mean	Std. Dev.	Min	Max
age	38.44	13.13	17	90
Education_num	10.12	2.55	1	16
Hours_per_week	40.93	11.98	1	99
Capital_gain	1092.01	7406.35	0	99999
Capital_loss	88.37	404.30	0	4356

Note that all numerical values reported in Table 1 are computed from the cleaned dataset used in the empirical analysis.

3 Exploratory Data Analysis

This section presents a descriptive analysis of the dataset. The goal is to better understand the main characteristics of the variables and their relationship with income. Both numerical and categorical variables are examined using graphical tools and summary statistics. The analysis is purely exploratory and serves as a foundation for the modeling approaches used later in the project.

3.1 Numerical Variables: Distributions

The distribution of age shows a concentration between roughly 25 and 55 years, with a right-skewed tail reflecting fewer older individuals. This pattern is consistent with a working-age population.

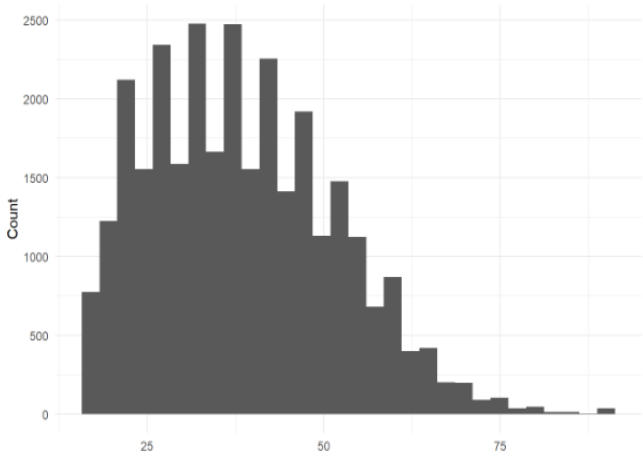

Figure 1: Histogram of age

Education level, measured by **education_num**, takes discrete values and is concentrated around middle levels, indicating that most individuals have completed secondary education or some college. Very low and very high education levels are less frequent.

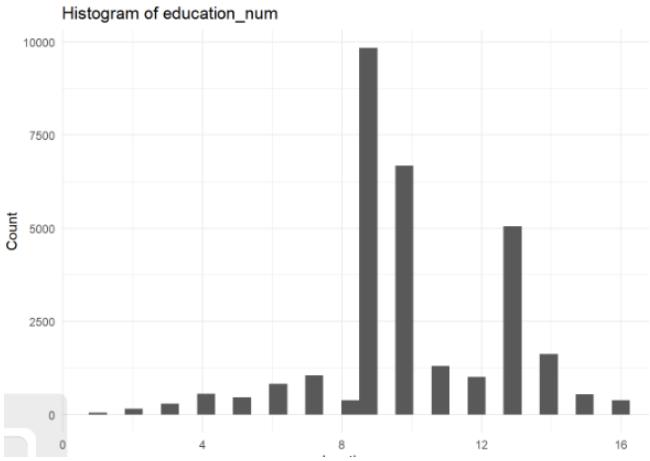

Figure 2: Histogram of education_num

Hours worked per week are strongly concentrated around 40 hours, reflecting standard full-time employment. A smaller number of individuals report very long working hours, which creates a right-skewed distribution.

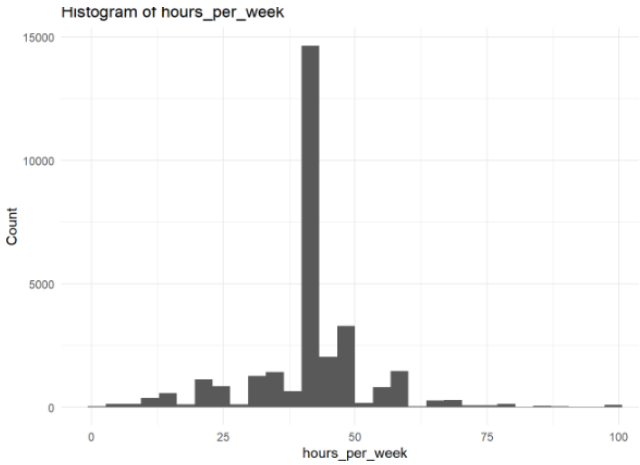

Figure 3: Histogram of hours_per_week

Both **capital_gain** and **capital_loss** display extreme right skewness. The vast majority of observations are equal to zero, while positive values occur rarely but can be very large. This distributional feature motivates the use of log-transformed variables for visualization.

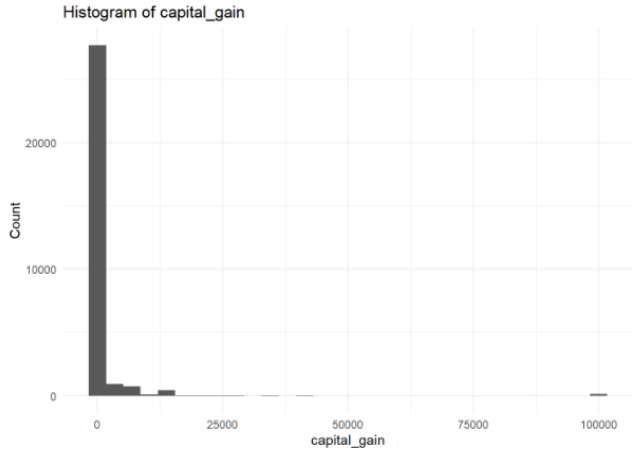

Figure 4: Histogram of capital_gain

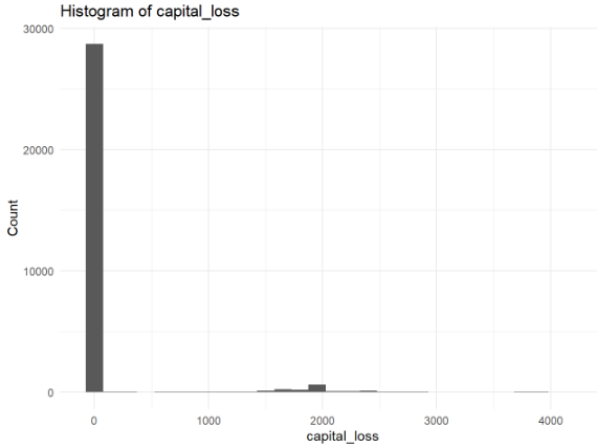

Figure 5: Histogram of capital_loss

3.2 Numerical Variables by Income Group

To better understand income differences, selected numerical variables are compared across income groups using boxplots. The focus is placed on variables that show clear and informative contrasts between individuals earning more than 50K and those earning 50K or less.

Education level, measured by **education_num**, differs strongly across income groups. Individuals earning more than 50K generally have higher education levels, with a noticeably higher median value. Although there is some overlap between groups, education appears to be an important factor associated with higher income.

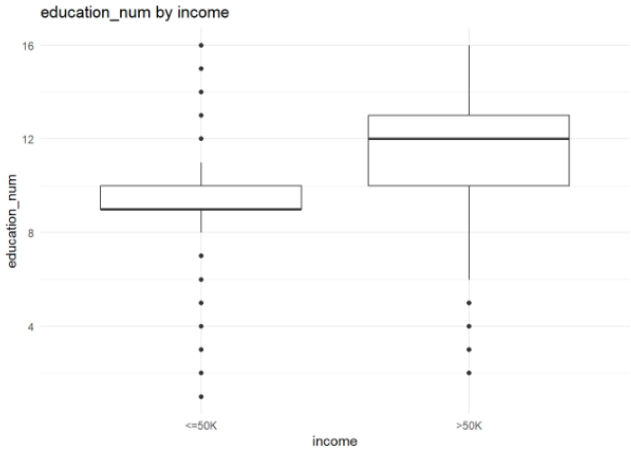

Figure 6: Education_num by income

Weekly **working hours** also vary by income group. Individuals in the higher-income category tend to work more hours per week on average. At the same

time, both income groups display substantial dispersion, indicating that working time alone does not fully explain income differences.

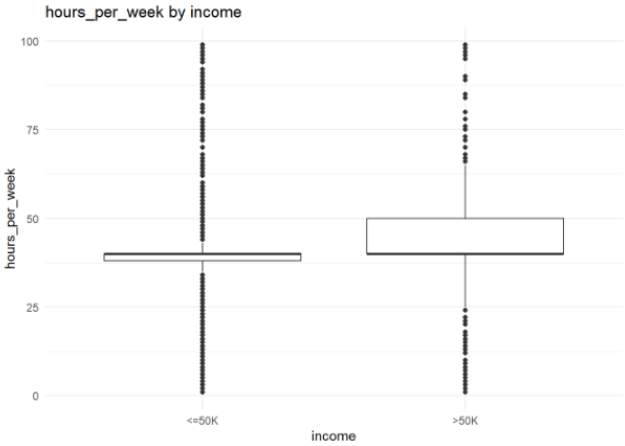

Figure 7: hours per Week by income

Capital gains show a particularly strong contrast between income groups. After applying a logarithmic transformation, positive values of **capital_gain** are observed much more frequently among individuals earning more than 50K. In contrast, most individuals in the lower-income group report no capital gains at all. This suggests that capital income plays a relevant role in distinguishing income levels.

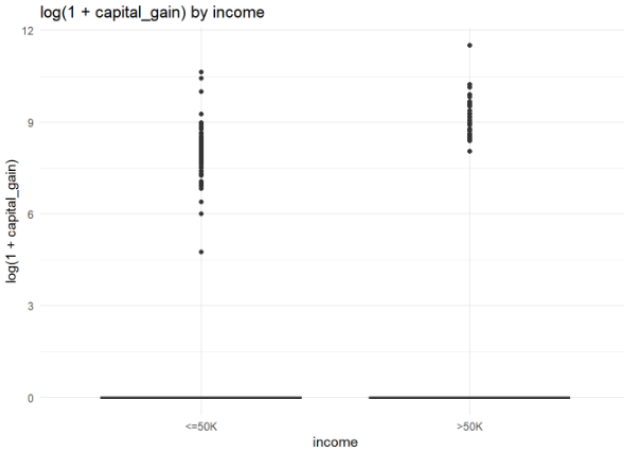

Figure 8: capital_gain by income

Additional boxplots for **age** and **capital_loss** were examined but are less informative. While higher-income individuals tend to be slightly older on average, the overlap between groups remains large. Capital losses are rare in both income groups and therefore provide limited additional information for income differentiation

3.3 Categorial Variables and Income Share

Income differences are further explored using categorical variables. Barplots are used to compare the proportion of individuals earning more than 50K across different groups. All plots are presented in relative terms, allowing for meaningful comparison across categories of different sizes.

Clear differences are observed across **sex**. The share of individuals earning more than 50K is noticeably higher among males than among females. For both groups, the majority of individuals earn 50K or less, but the gap between income categories is more pronounced for males.

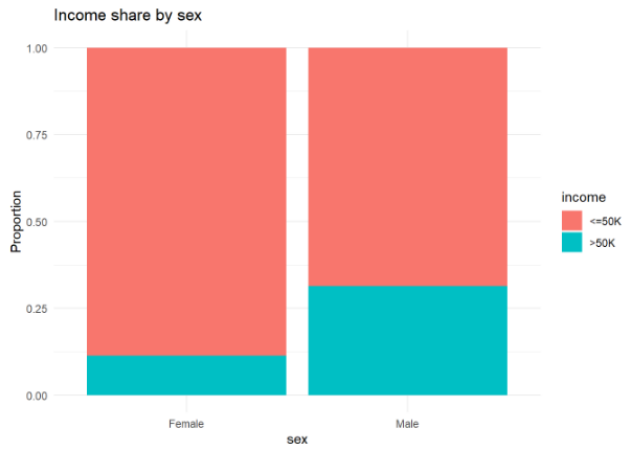

Figure 9: Income share by sex

Income composition also varies across **race**. While most racial groups are dominated by the lower-income category, some groups show a relatively higher share of individuals earning more than 50K. At the same time, no group is composed primarily of high-income earners, indicating substantial overlap across racial categories.

Differences are particularly strong for **marital_status**. Married individuals, especially those married to a civilian spouse, display a much higher proportion of incomes above 50K. In contrast, never-married, separated, and divorced individuals are largely concentrated in the lower-income group. This suggests that marital status is strongly associated with income outcomes in the dataset.

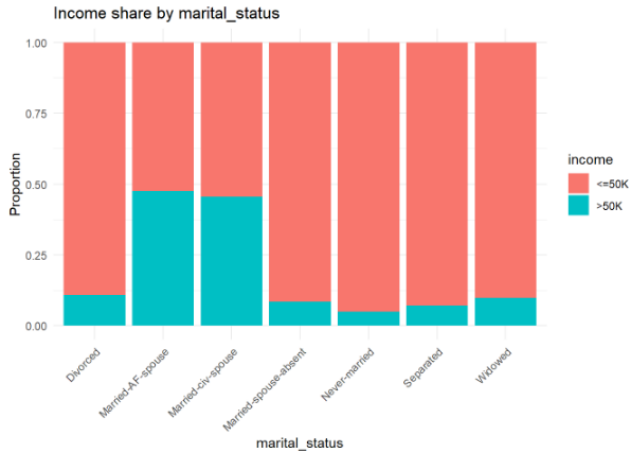

Figure 10: Income share by marital_status

Income shares also differ across **workclass** categories. Individuals who are self-employed or working in the private sector tend to have a higher proportion of high-income earners compared to other work classes. In contrast, work classes such as without-pay or government-related categories show very low shares of incomes above 50K.

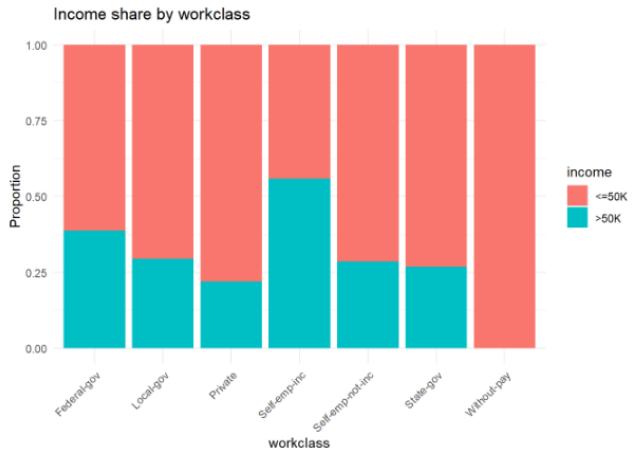

Figure 11: Income share by workclass

Overall, these barplots highlight substantial heterogeneity in income outcomes across categorical variables. While none of the variables perfectly separates income groups, several show clear differences in income shares. These patterns motivate the inclusion of categorical variables in subsequent modeling and hypothesis testing.

3.4 Correlation Structure of Numerical Variables

To explore linear relationships between numerical variables, a correlation heatmap is used. The analysis focuses on age, education_num, hours_per_week, capital_gain, and capital_loss.

Overall, most pairwise correlations are weak. This indicates that the numerical variables are not strongly linearly related to each other. In particular, age shows only limited correlation with education and working hours.

Education level displays moderate positive correlation with both hours worked per week and capital gains. This suggests that higher education is associated with longer working hours and a higher likelihood of receiving capital income, although the relationships remain far from perfect.

Capital gain and capital loss show very low correlations with most other variables. This reflects the fact that these variables contain many zero values and only a small number of positive observations.

In summary, the correlation analysis suggests that no single numerical variable dominates the data. Instead, income-related patterns are likely driven by a combination of several variables rather than strong linear relationships between individual covariates.

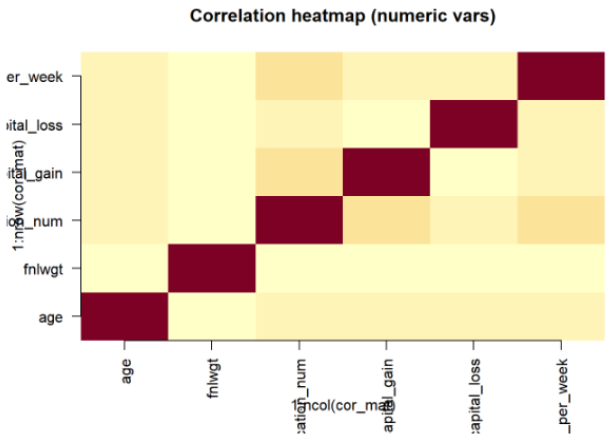

Figure 12: correlation heatmap (num variables)

3.5 Summary of Exploratory Findings

The exploratory data analysis highlights several important patterns in the Adult dataset. Income differences are associated with education level, working hours, and capital income, while no single variable fully separates high- and low-income individuals. Numerical variables show substantial dispersion, especially capital gains and losses, which are highly skewed.
Categorical variables such as sex, marital status, and workclass display clear differences in income shares across groups, although considerable overlap remains. Overall, the exploratory analysis suggests that income inequality reflects the combined effect of multiple socio-economic factors rather than simple linear relationships. These findings motivate the use of supervised learning models for prediction and unsupervised methods to explore latent heterogeneity in the data.

4 Methodology

This section describes the methodological approach used in the empirical analysis.
The objective is not causal inference, but to **analyze income inequality**, **evaluate predictive performance**, and **explore latent socio-economic structures** in the Adult dataset.
The analysis combines **supervised learning methods**, which use income as an outcome variable, and **unsupervised learning methods**, which identify groups without using income. All models are implemented in R using consistent preprocessing and evaluation procedures.

4.1 Supervised Learning

Supervised learning methods are applied to predict whether an individual earns more than $50K per year. The outcome variable is the binary income indicator (`<=50K`, `>50K`).

4.1.1 Choice of Explanatory Variables

The supervised learning models are built using a subset of individual-level characteristics that are commonly associated with income determination. The explanatory variables retained for the analysis include **age, education_num, hours_per_week, capital_gain, capital_loss, sex, marital_status, race, and workclass**. These variables capture key

dimensions of human capital, labor supply, demographic structure, and occupational status, and are widely used in empirical studies of income inequality and wage outcomes.

Other variables available in the dataset are deliberately excluded. In particular, **fnlwgt** is not used because it represents a sampling weight rather than an intrinsic characteristic of individuals and is therefore not suitable as a predictor in classification models. The variable **native_country** is omitted due to its high number of categories and its limited additional contribution to predictive performance in preliminary analyses. Finally, variables that are highly correlated or redundant are avoided in order to preserve model stability, interpretability, and comparability across specifications.

4.1.2 Logistic Regression (GLM)

Logistic regression is employed as the primary supervised learning method in the analysis. The model estimates the probability that an individual earns more than $50K per year as a function of the selected explanatory variables. This approach is well suited to the binary nature of the income outcome and provides a clear probabilistic interpretation of predictions.

Logistic regression is chosen for several reasons. First, it offers a high degree of interpretability, allowing estimated coefficients to be directly linked to changes in the likelihood of belonging to the high-income group. Second, the model produces predicted probabilities rather than only class labels, which is useful for understanding uncertainty in classification decisions. Finally, logistic regression serves as a standard benchmark in binary classification tasks, making it an appropriate reference point for model comparison.

Model performance is assessed using a train–test split to evaluate out-of-sample predictive ability. Classification accuracy and confusion matrices are used to summarize overall performance and to examine the balance between correct and incorrect classifications across income groups.

4.1.3 k-Nearest Neighbors (kNN)

As a comparison model, a **k-nearest neighbors (kNN)** classifier is implemented.

This method classifies individuals based on similarity in the standardized feature space, without assuming a parametric relationship between predictors and income.

Before applying kNN, all numeric variables are standardized to ensure that distance calculations are not dominated by variables with large scales (e.g., capital gains).
kNN is included to assess whether a **non-parametric approach** improves prediction, and to compare performance with the more interpretable logistic regression model.

4.1.4 Model Comparison

The predictive performance of logistic regression and kNN is compared using out-of-sample accuracy.
The model with the highest test accuracy is selected as the final supervised learning model.

4.2 Unsupervised Learning

Unsupervised learning methods are used to explore whether individuals form **natural socio-economic groups** based on observable characteristics. Importantly, the income variable is **not used** during clustering to avoid information leakage.

4.2.1 Choice of Clustering Variables

Clustering is performed using the variables age, education_num, hours_per_week, capital_gain, and capital_loss. These variables are selected because they capture key dimensions of income formation, including human capital, labor supply, and capital income, which are central to the study of income inequality. Together, they provide a concise yet informative representation of individual economic profiles.
Other variables are intentionally excluded from the clustering analysis. In particular, categorical variables such as race and workclass are not used because k-means clustering relies on Euclidean distance and is therefore not well suited to non-numeric features. In addition, the income variable itself is excluded to ensure that clustering remains purely exploratory and does not mechanically reproduce the observed income groups. Prior to clustering, all selected variables are standardized to have zero mean and unit variance in order to prevent differences in scale from dominating the distance calculations.

4.2.2 Choice of the Number of Clusters

To determine the appropriate number of clusters, the elbow method is applied. This method examines how the within-cluster sum of squares (WSS) changes as the number of clusters increases. As the number of clusters grows, WSS naturally decreases, but beyond a certain point the improvement becomes marginal. This point is referred to as the "elbow" and provides guidance for selecting K.

To reduce subjectivity in this choice, an automatic elbow detection rule is used in addition to visual inspection. Both the elbow plot and the automatic procedure point to the same solution, indicating that K = 5 provides a reasonable balance between model simplicity and explanatory power.

At this stage of the analysis, only one figure is required. The elbow plot showing the relationship between K and the within-cluster sum of squares is included here to illustrate the selection of the number of clusters.

4.2.3 k-means Clustering

Based on the chosen number of clusters, k-means clustering is applied to the standardized data. Each individual is assigned to a cluster according to their characteristics. Income is not used during the clustering process but is introduced afterward to interpret the results. This makes it possible to examine how income levels differ across the identified socio-economic groups.

4.3 Reproducibility

All analyses are conducted using the R programming language. Identical preprocessing steps and evaluation procedures are applied across models to ensure transparency and reproducibility. All results reported in the paper are generated directly from the analysis code.

5 Results

This section presents the main empirical results from the supervised and unsupervised learning analyses. The focus is on predictive performance, comparison across models, and interpretation of the identified patterns. No causal interpretation is intended.

5.1 Supervised Learning Results

5.1.1 Classification Models

The main supervised learning task is the prediction of whether an individual earns more than $50K per year. Two classification models are considered: logistic regression (GLM) and k-nearest neighbors (kNN).

Table 2 reports the out-of-sample performance of both models based on a common train–test split.

Table 2: classification performance

Model	Accuracy
Logistic regression (GLM)	0.846
kNN (k = 15)	0.830

In addition to accuracy, the F1-score was computed to account for class imbalance between income groups. The F1-score confirms that the logistic regression model achieves a good balance between correctly identifying high-income individuals and limiting false classifications.
Overall, logistic regression achieves the highest accuracy and is therefore selected as the preferred model. Although the difference in performance between the two methods is moderate, it is consistent. This suggests that income status can be well captured by a relatively small set of key predictors, and that a global parametric model performs slightly better than a local distance-based approach in this application.
Given its superior predictive performance, the logistic regression model is examined in more detail below.

5.1.2 Interpretation of the Logistic Regression Model

The interpretation of the logistic regression results is based on the estimated coefficients, their signs, and their associated p-values. A positive coefficient indicates that an increase in the corresponding variable raises the probability of earning more than $50K. Statistical significance, assessed

through p-values, indicates whether this relationship is unlikely to be due to random variation once other covariates are controlled for.

The results show that education level and labor supply play a central role in income determination. In particular, *education_num* and *hours_per_week* have positive and highly statistically significant coefficients, with very small p-values. This implies that higher educational attainment and longer working hours significantly increase the probability of earning more than $50K.

Capital income variables, namely *capital_gain* and *capital_loss*, are also strongly statistically significant. Their low p-values indicate a robust association between access to non-labor income and high-income status, highlighting the importance of capital income in explaining income inequality.

Several demographic characteristics, including *sex* and *marital_status*, remain statistically significant at conventional significance levels. In contrast, some categories related to *race* and *workclass* exhibit larger p-values, indicating weaker or statistically insignificant effects once education, labor supply, and capital income are taken into account.

Overall, the logistic regression results suggest that income inequality in the dataset is primarily driven by differences in human capital, working intensity, and access to capital income, while purely demographic characteristics play a more limited role once economic factors are controlled for.

5.2 Unsupervised Learning Results

5.2.1 Choice of the Number of Clusters

Unsupervised learning is used to identify groups of individuals with similar socio-economic characteristics without using income information. K-means clustering is applied to standardized continuous variables related to human capital, labor supply, and capital income.

The number of clusters is selected using the elbow method. The elbow plot shows a clear reduction in within-cluster variation up to **K = 5**, after which additional clusters lead to only small improvements. Based on this criterion, **five clusters are retained** for further analysis. This choice represents a balance between model simplicity and the ability to capture heterogeneity in the data.

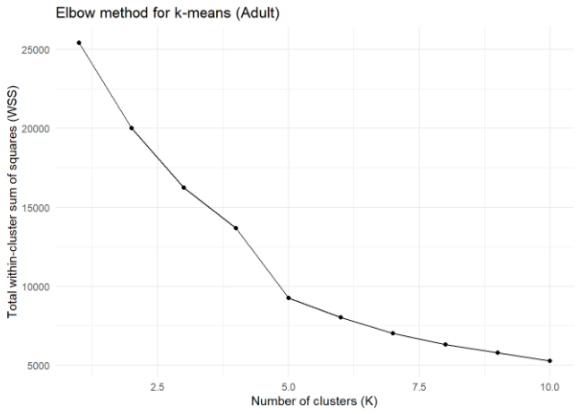

Figure 13: Elbow method for K-means (Adult)

5.2.2 Cluster Structure and Profiles

After fixing the number of clusters at K = 5, k-means clustering is applied to the full standardized dataset. Each individual is assigned to one cluster based on similarities in age, education level, working hours, and capital income variables.

The resulting clusters differ substantially in size and socio-economic composition. Some clusters are characterized by higher education levels and longer working hours, while others consist mainly of individuals with lower education and weaker labor market attachment. One small cluster is dominated by individuals with very high capital gains, clearly separating them from the rest of the population.

These differences suggest that the clustering procedure successfully identifies distinct socio-economic profiles rather than random groupings.

5.2.3 Relationship Between Clusters and Income

Although income is not used during the clustering process, it is introduced afterward to interpret the clusters. Clear differences in income distribution are observed across clusters. Some clusters contain a high proportion of individuals earning more than $50K, while others are largely composed of low-income individuals.

In particular, clusters characterized by higher education, longer working hours, or substantial capital income exhibit much higher shares of high-income individuals. In contrast, clusters with lower education levels and

shorter working hours are strongly concentrated in the low-income category.
These results indicate that the clusters capture meaningful socio-economic structures that are closely related to income inequality, even though income itself was excluded from the clustering stage.

5.3 Post-Clustering Interpretation

Post-clustering analysis is used to better understand the socio-economic meaning of the clusters identified by the k-means algorithm. Although income is not used to form the clusters, it provides a useful reference for interpreting differences across groups.
The analysis reveals clear and systematic differences in economic outcomes across clusters. Clusters characterized by higher average education levels, longer working hours, and positive capital income display a much higher proportion of individuals earning more than $50K. In contrast, clusters composed of individuals with lower education levels, shorter working hours, and little or no capital income are largely concentrated in the low-income category.
One cluster stands out due to the presence of very high capital gains, with nearly all individuals in this group belonging to the high-income category. This cluster highlights the importance of capital income as a key dimension of income inequality that is not captured solely by labor market characteristics.
Beyond income, clusters also differ in terms of age composition. Some clusters consist mainly of younger individuals with relatively low income and limited capital income, while others include older individuals with more accumulated resources. These patterns suggest that income inequality reflects both differences in labor market outcomes and life-cycle effects.
Overall, the clustering results indicate that unsupervised learning uncovers meaningful socio-economic profiles that align closely, though not perfectly, with income differences. This confirms the value of clustering as a complementary tool to supervised learning, providing additional insight into the structure of inequality beyond direct income prediction.

Table 3:Distribution of income categories across k-means clusters (K = 5)

	<=50K	>50K
1	47.3	52.7
2	56.6	43.4

3	0.0	100.0
4	92.8	7.2
5	76.8	23.2

6 Discussion

This study applies supervised and unsupervised learning methods to income data in order to analyze predictive performance, interpretability, and income inequality patterns. The discussion below directly builds on the numerical results obtained in the empirical analysis and highlights their economic and methodological implications.

6.1 Predictive Performance and Model Choice

In the supervised learning framework, two classification models are compared to predict whether an individual earns more than $50K per year: logistic regression (GLM) and k-nearest neighbors (kNN). Based on the test set results, logistic regression achieves an accuracy of **84.6%**, while kNN reaches **83.0%**.

Although the difference in accuracy is relatively small (around **1.6 percentage points**), logistic regression consistently performs better and is therefore selected as the preferred model. This result suggests that income status can be well explained by a limited number of strong predictors and does not require highly flexible local methods to achieve good predictive performance.

The comparison highlights an important trade-off. More complex models such as kNN do not provide substantial gains in accuracy, while logistic regression offers clearer interpretation and probabilistic predictions. In the context of income analysis, model transparency and interpretability are therefore important advantages.

6.2 Interpretation of Income Determinants

The logistic regression results provide clear insights into the determinants of income inequality. Education level and labor supply play a central role in explaining high-income status. In particular, **education_num** and **hours_per_week** have positive coefficients and are highly statistically significant, with p-values smaller than **0.05**. This indicates that individuals with higher education and longer working hours are significantly more likely to earn more than $50K.

Capital income variables, namely **capital_gain** and **capital_loss**, are also strongly significant. Their very small p-values indicate that non-labor income is an important factor in distinguishing high-income individuals. This highlights that income inequality in the dataset is not driven solely by labor market outcomes, but also by unequal access to capital income.

Several demographic variables, such as **sex** and **marital_status**, remain statistically significant even after controlling for education and working hours. In contrast, some categories related to **race** and **workclass** are not statistically significant once the main economic variables are included. This suggests that part of the observed demographic income gaps reflects underlying differences in education, labor supply, and capital income rather than independent demographic effects.

Overall, the logistic regression results confirm that income inequality in the dataset is primarily associated with differences in human capital, working intensity, and access to capital income.

6.3 Insights from Unsupervised Learning

Unsupervised learning is used to explore latent group structures without using income information. K-means clustering, applied to standardized variables related to age, education, working hours, and capital income, automatically selects **five clusters** using the elbow method.

The resulting clusters display substantial heterogeneity in their average characteristics. When income is introduced only for interpretation, strong differences in income composition across clusters emerge. For example, one cluster consists exclusively of high-income individuals (**100% earning more than $50K**), while another cluster is dominated by low-income individuals (**over 90% earning $50K or less**). Other clusters show more mixed income distributions.

At the same time, clusters do not perfectly correspond to income categories. Each cluster contains individuals from both income groups, indicating

considerable within-cluster variation. This confirms that income inequality cannot be reduced to a small number of homogeneous socio-economic groups.
Unsupervised learning therefore complements the supervised analysis by revealing broader socio-economic profiles rather than providing direct income predictions.

6.4 Methodological Implications

The results illustrate the complementary roles of supervised and unsupervised learning methods. Supervised models, especially logistic regression, are well suited for prediction and for identifying key income determinants. Unsupervised methods, such as k-means clustering, provide exploratory insights into heterogeneity but should not be interpreted as predictive or causal tools.
Logistic regression proves to be a strong baseline model, combining good predictive accuracy with clear interpretability. Clustering methods are useful for descriptive analysis but require cautious interpretation, as clusters only partially align with observed income outcomes.

6.5 Limitations

Several limitations should be acknowledged. The analysis relies on observational data, which prevents causal interpretation of the estimated relationships. Important factors such as job quality, firm characteristics, or career dynamics are not observed. In addition, the results are specific to the dataset and may not generalize to other labor markets or time periods.
Despite these limitations, the study provides a coherent and transparent application of statistical learning methods to income data and demonstrates how different approaches can be combined to better understand income inequality.

7 Conclusion

This study applies supervised and unsupervised statistical learning methods to income data in order to analyze income inequality and compare different modeling approaches. The analysis focuses on prediction performance, interpretability, and exploratory insights rather than on causal inference.
The supervised learning results show that income status can be predicted with good accuracy using a limited number of key variables. Logistic

regression performs slightly better than k-nearest neighbors and is therefore selected as the main classification model. Education level, working hours, and capital income variables consistently emerge as the most important and statistically significant predictors. These results highlight the central role of human capital, labor supply, and access to capital income in explaining income inequality.

Unsupervised learning provides complementary insights. K-means clustering identifies several distinct socio-economic profiles based on age, education, working hours, and capital income. When income is introduced only for interpretation, clear differences in income composition across clusters appear. However, clusters do not perfectly match income categories, indicating that income inequality cannot be fully captured by a small number of homogeneous groups.

From a methodological perspective, the study illustrates the value of combining exploratory data analysis, probabilistic reasoning, supervised learning, and clustering within a unified framework. Supervised models are well suited for prediction and interpretation, while unsupervised methods are useful for exploring heterogeneity in the data. Overall, the project provides a clear and reproducible example of how statistical learning methods can be applied to income data and inequality analysis.

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